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Research Article

Backdoor Attack and Defense Methods for AI–Based IoT Intrusion Detection System

Bowen Ma,[1] **Jiangwei Shi**,[2] **Ning Zhu**,[1] **Chen Fang**,[1] **and Yongjin Hu**[1]

[1]*People's Liberation Army Strategic Support Force Information Engineering University, Zhengzhou, Henan, China*
[2]*Air Force Engineering University, Xi'an, Shaanxi, China*

Correspondence should be addressed to Jiangwei Shi; 2016122107@chd.edu.cn

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The Internet of Things (IoT) is an emerging technology that has attracted significant attention and triggered a technical revolution in recent years. Numerous IoT devices are directly connected to the physical world, such as security cameras and medical equipment, making IoT security a critical issue. Artificial intelligence (AI) based intrusion detection technology for IoT can rapidly detect network attacks and improve security performance. However, this technology is vulnerable to backdoor attacks. As an important form of adversarial machine learning (ML), backdoor attacks can allow malicious traffic to evade detection of the intrusion detection system, posing a significant threat to the IoT security. This study focuses on backdoor attack and defense methods for AI–based IoT intrusion detection system. Specifically, we first use different ML and deep learning (DL) classification models to classify IoT traffic data, thereby achieving intrusion detection within IoT. Additionally, we employ data poisoning techniques to implant backdoors into models, enabling backdoor attacks on classification models. For backdoor defense, we propose backdoor detection and mitigate methods: (1) The proposed backdoor detection method is achieved by leveraging the strong correlation between the backdoor trigger and the target classification; (2) we utilize the unlearning method to mitigate the backdoor effect, enhancing the robustness of classification networks. Extensive experiments were conducted on the CICIOT2023 dataset to evaluate the effectiveness of IoT intrusion detection, backdoor attack, and defense.

Keywords: backdoor attack; backdoor defense; Internet of Things; intrusion detection system

1. Introduction

The Internet of Things (IoT) is a rapidly evolving technology that has garnered substantial research interest and undergone significant technological advancements in recent years [1]. IoT plays a crucial role in many smart applications such as smart grids, smart homes, wearable devices, environmental monitoring, robotics, and vehicle networks [2]. IoT technology optimizes the use of limited resources, increasing industrial efficiency and profitability. However, as IoT devices exchange vast amounts of data with external resources, attackers may infiltrate IoT systems that lack appropriate security measures, causing severe damage. Unlike traditional internet systems, IoT devices are often low-cost, with limited computing capabilities, and exhibit global connectivity, temporality, and mobility. These characteristics lead to inadequate security measures, requiring continuous monitoring and analysis to ensure safety [3, 4]. Thus, analyzing and detecting network attacks targeting IoT has become a hot research topic.

Network intrusion detection systems (NIDSs) monitor and analyze network traffic to detect potential attacks. Research on NIDS began in the 1980s and has since become an integral part of network security architecture [5]. However, as digital technology advances and cyberattacks diversify, the complexity of network traffic data increases, making traditional NIDS insufficient for accurately identifying anomalies and unknown attacks in IoT. With the rapid development of artificial intelligence (AI), particularly the advancements in machine learning (ML), has significantly improved the performance and efficiency of intrusion detection systems. For instance, Zhang et al.

[6] achieved a 95% classification accuracy for intrusion detection using support vector machine (SVM). Adetunmbi et al. [7] proposed a *K*-nearest neighbors (KNNs)–based IoT anomaly and intrusion detection classification method, improving overall performance through feature reduction. As IoT data traffic grows, deep learning (DL)–based intrusion detection methods have garnered increasing attention. DL methods can capture complex data patterns, achieving high precision in large-scale nonlinear datasets. For example, Wang et al. [8] proposed an recurrent neural network (RNN)–based anomaly detection method for industrial control systems in the power sector, analyzing application-layer payload data and achieving high detection success rates. As AI technologies are increasingly integrated into various practical applications, the issue of robustness has become a critical challenge that cannot be ignored. Attacks such as adversarial example attacks, poisoning attacks, and backdoor attacks, which target AI models directly, can compromise the confidentiality, integrity, and availability of models and pose significant threats to AI–based security systems [9, 10].

Backdoor attacks [11] target AI models, where attackers implant backdoors during the training process, causing the model to misclassify samples with specific triggers as a target class while correctly classifying clean samples. In real-world applications, when the model training process is not fully controlled (e.g., outsourced training or using third-party pretrained models), attackers can implant backdoors through poisoned data or model parameter modifications, causing the model to predict according to the attacker's intentions. Since backdoor models only activate with specific triggers known exclusively to the attacker and perform normally on clean samples, this attack method is highly concealed.

Backdoor attacks were first introduced by Gu et al. [12] in computer vision (CV), where attackers successfully implanted backdoors into a traffic sign detection model, causing it to misidentify a "Stop" sign with an added trigger as a "Speed Limit" sign, demonstrating the significant threat posed by backdoor attacks. Since then, backdoor attacks have been widely studied in areas such as natural language processing (NLP), graph neural networks, reinforcement learning, and federated learning, posing serious threats across multiple domains [13]. In network security, some studies have used backdoor attacks to target malware detection systems and traffic classification systems. Severi et al. [14] created backdoor triggers in a model-agnostic manner, successfully attacking classifiers for various applications, including Windows PE files, PDF files, and Android applications. Ning et al. [15] successfully executed backdoor attacks on traffic classifiers, achieving an attack success rate (ASR) of up to 98.3%. Unlike traditional internet networks, IoT traffic is characterized by being small-scale and high-frequency, and with the continuous emergence of new attack methods, and AI–based IoT intrusion detection systems are vulnerable to backdoor attacks, primarily due to the following factors: First, the inherent resource constraints of IoT devices limits their ability to deploy computationally intensive defense mechanisms, resulting in a heightened reliance on individual AI models. Consequently, if a backdoor attack successfully bypasses such a model, it can lead to severe and potentially undetected malicious behavior, posing a significantly greater risk to IoT environments compared to traditional IT systems, which typically incorporate multilayered and redundant security architectures. Second, the large-scale distributed nature of IoT systems complicates security updates and patch management, providing opportunities for backdoor attacks, additionally, the data transmission process in IoT devices is susceptible to man-in-the-middle attacks, which increases the risk of backdoor implantation. Finally, many IoT devices utilize standardized hardware and open-source software, allowing attackers to exploit known vulnerabilities or compromise the supply chain to implant backdoors. These characteristics collectively make IoT intrusion detection systems a prime target for backdoor attacks. According to our review, there has been no prior work dedicated to examining backdoor attacks and their respective defenses in the realm of AI–based IoT intrusion detection systems. Studying backdoor attacks and defenses can further enhance our understanding of the vulnerabilities in IoT intrusion detection systems, improving their robustness.

In this study, we focus on how to leverage backdoor attacks to target AI–based IoT intrusion detection systems and how to defend against such attacks. Specifically, backdoor samples are created by embedding triggers into a portion of the traffic samples, which are then mixed into the clean dataset. The mixed data is used to train classification models, thereby implanting the backdoor. Malicious traffic with triggers is then used to attack the classification model to evaluate the effectiveness of the backdoor attack. More importantly, we propose a combined defense mechanism that integrates both backdoor detection and backdoor mitigation to resist backdoor attacks while improving the robustness of the classification networks. Backdoor detection is used to identify whether triggers exist in the data, while backdoor mitigation employs unlearning to retrain or fine-tune the backdoor model, reducing the impact of the backdoor and enhancing the robustness of the classification network.

The main contributions of this study are as follows:

1. Intrusion detection for the IoT has been implemented using six ML–based classifiers, including three ensemble learning-based classifiers: AdaBoost, gradient boosting decision tree (GBDT), and eXtreme gradient boosting (XGBoost) as well as three DL–based classifiers: multilayer perceptron (MLP), convolutional neural network (CNN), and RNN. The results obtained from these methods have shown significant success in detection, the detection success rate for each of the aforementioned classifiers exceeds 91.82%.
2. A backdoor attack method has been designed to successfully carry out backdoor attacks on IoT intrusion detection systems that rely on AI. Extensive experiments have been conducted on the CICIOT2023 dataset, revealing that malicious traffic embedded with triggers could be classified as benign by the backdoor-compromised classification network, achieving an ASR of over 98.44%. Meanwhile, it has been observed that both clean benign and malicious traffic continue to be detected accurately.

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This demonstrates the susceptibility of AI–based IoT intrusion detection systems to backdoor attacks.

3. Defense methods against backdoor attacks have been proposed from two aspects: detection and mitigation. Backdoor detection is achieved by leveraging the strong correlation between the trigger and the benign label, while unlearning is used to mitigate the backdoor effect. Evaluations conducted on the CICIOT2023 dataset demonstrate that the proposed defense methods reduce the ASR across six classifiers from over 97% to below 0.5%, indicating that the proposed method is effective in defending IoT intrusion detection systems against backdoor attacks.

The rest of this study is organized as follows: Section 2 in brief summarizes related work in the field. Section 3 introduces the classification of IoT traffic, backdoor attacks, and backdoor defenses. A series of experimental results and analyses are presented in Section 4. Finally, Section 5 provides the conclusion.

2. Related Work

2.1. AI–Based Intrusion Detection System for IoT. For IoT intrusion detection, common methods include signature-based detection and anomaly-based detection, signature-based detection relies on preestablished signatures or knowledge of attacks, making it incapable of detecting unknown threats. In contrast, anomaly-based detection works by learning and establishing a model of "normal" behavior, thereby identifying potential attacks or anomalies that deviate from this norm. The rapid advancement in AI has significantly propelled the development of anomaly-based intrusion detection, AI–based anomaly detection methods [16] can automatically learn new features and patterns from training data, offering better adaptability and predictive capabilities.

In terms of ML, traditional algorithms such as decision tree (DT), random forests (RFs), logistic regression (LR), and Naive Bayes (NB) have achieved good results in IoT intrusion detection [17]. However, these algorithms often lack generalization capability and are insufficient for handling complex tasks. In contrast, ensemble learning algorithms like Adaboost, GBDT, and XGBoost combine multiple learners to improve detection performance, showing significant improvements in accuracy, generalization, and high-dimensional data processing [18]. For example, Zhou et al. [19] proposed an ensemble intrusion detection system based on Adaboost, which improved recognition and classification performance of different intrusion behaviors through optimized sampling ratios, DT depth, and iteration times, achieving a detection rate of 99% on the AWID wireless network dataset. Cui et al. [20] proposed an optimized GBDT intrusion detection algorithm to protect edge computing centers in the industrial IoT, improving detection efficiency and success rate. Doghramachi and Ameen [21] combined XGBoost with SMOTE to detect attacks in IoT datasets, showing significant improvements in attack detection on imbalanced IoT datasets.

As the volume of data increases, DL algorithms such as MLP, CNN, and RNN have become prominent in handling large-scale, high-dimensional, and nonlinear datasets [22]. MLPs, being multilayer neural networks, excel at learning complex patterns and features from large amounts of data. Yang et al. [23] combined autoencoders with MLPs to enhance the detection of imbalanced attacks by learning robust input features, yielding better detection performance. CNNs, on the other hand, have powerful spatial feature extraction capabilities through convolutional operations, making them widely applicable in intrusion detection. For example, Garg et al. [24] used gray wolf optimizer (GWO) and CNN techniques to develop a robust hybrid model for anomaly detection in cloud environments, achieving improvements in detection accuracy through an optimized GWO and CNN model. In fog-based IoT environments, N.G. and S. [25] introduced vector convolution to build an intrusion detection system, with a distributed DL approach to handle scalable data. RNNs, known for their sequential data processing capabilities, effectively capture temporal information. In intrusion detection, viewing network traffic as a sequence of events, RNNs can effectively identify network attacks. Imrana et al. [26] enhanced the detection of various attacks using RNNs, significantly improving detection performance. Goh et al. [27] used RNNs to detect anomalies in cyber-physical systems (CPSs) and identify attacked sensors, demonstrating effective detection of most attacks with low false positive rates (FPRs).

2.2. Backdoor Attacks and Defenses. As ensemble learning and DL algorithms are increasingly applied to IoT intrusion detection systems, the inherent robustness issues of these models introduce new security threats. From an attack perspective, the main threats targeting model accuracy include poisoning attacks, adversarial example attacks, and backdoor attacks. Poisoning attacks degrade the generalization performance of models, preventing them from making correct predictions [28, 29]. Adversarial example attacks deceive models by constructing adversarial samples that cause incorrect predictions [30, 31]. Backdoor attacks, however, force the model to learn attacker-specified triggers, producing normal predictions for clean samples while outputting attacker-designated results for triggered samples [32, 33]. Compared to adversarial example attacks, which occur during the inference phase, backdoor attacks typically happen during the model training phase. The model training process involves multiple steps, including data collection, data preprocessing, model construction, model saving, and model deployment. More steps mean more opportunities for attackers to intervene, thereby increasing the attack surface of the model. Compared to poisoning attacks, which also occur during the training phase, backdoor attacks demonstrate better stealth. They maintain normal detection performance on clean samples, only exhibiting anomalous behavior when encountering trigger-embedded samples, making them difficult to detect during testing or operation. In contrast, poisoning attacks directly degrade model accuracy, making them more likely to be detected during user testing.

Backdoor attacks against ML were first proposed in the CV domain [12], since then, backdoor attacks and their defenses have been extensively studied across various fields. For backdoor attacks, in CV, Li et al. [34] used image steganography and

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the least significant bit algorithm to create invisible triggers, which could not be detected by humans, but could still activate the model's backdoor. In NLP, Pan et al. [35] proposed a backdoor attack method based on language style, which utilizes implicit language style as a hidden trigger for poisoning NLP models, achieved attack effectiveness by preserving malicious semantics, while also ensuring stealthiness in terms of the naturalness of the trigger and evasion of detection. In the field of reinforcement learning, Wang et al. [36] demonstrated the feasibility of backdoor attacks in deep reinforcement learning systems, the adversary agent can trigger the backdoor of the victim agent through its own actions, resulting in a decrease in the victim's winning rate by 17%–37%. In terms of defense, existing backdoor defense methods mainly include backdoor detection, recognition, and suppression [11]. Backdoor detection aims to determine whether a model or sample contains a backdoor. For example, Chen et al. [37] proposed detecting poisoned samples by clustering activations, assuming that poisoned samples form small clusters within the target class. Gao et al. [38] overlaid various image patterns onto input images and observed the randomness of model predictions; if a model consistently predicted a specific class, it was likely influenced by a backdoor. Backdoor recognition methods aim to identify the neurons or patterns responsible for the backdoor. Wang et al. [39] suggested that backdoor triggers create "shortcuts" between samples and target labels, which can be identified through an optimized search. Backdoor suppression methods reduce the effect of backdoors or prevent their insertion during training. For example, Liu et al. [40] proposed pruning part of the inactive neurons for clean samples and then fine-tuning the pruned model to recover performance using a subset of benign samples.

Samples in fields such as CV and NLP typically have high-dimensional feature spaces, allowing for a relatively large number of perturbed features. In contrast, samples used in intrusion detection systems usually have lower dimensionality, and any perturbation must satisfy constraints related to traffic validity and malicious intent. This limits backdoor triggers to specific, constrained ranges, resulting in lower-dimensional backdoor features. As a result, techniques developed in CV and NLP domains often cannot be directly applied to intrusion detection systems. Furthermore, in CV and NLP applications, the primary objective of backdoor attacks is typically to cause misclassification on specific inputs. However, in the context of intrusion detection, attackers may aim to bypass detection mechanisms entirely or render the system ineffective. These differences in threat models suggest that defense mechanisms designed for other domains may not be effective against backdoor attacks specifically tailored for intrusion detection systems.

To target intrusion detection systems, Holodnak et al. [41] combined the concept of universal adversarial perturbations with backdoor attacks, constructing poisoned data by padding network packets, achieving a backdoor ASR of over 90% on multiple target classes. Severi et al. [42] introduced explainable ML techniques into backdoor attacks, achieving significant ASRs with low poisoning rates across various network traffic datasets, including malware detection and application classification. On the defense side, Bachl et al. [43] transplanted methods from CV to defend against backdoor attacks in intrusion detection systems by pruning DT and RF, demonstrating effectiveness on two network security datasets. Despite these studies, research on backdoor attacks and defenses in traffic classification, especially for IoT intrusion detection systems, remains in its infancy, leaving much work to be explored.

3. Backdoor Attack and Defense Method for IoT Intrusion Detection Systems

With the rapid development of IoT technology, the amount of data generated by IoT devices is increasing exponentially. At the same time, ML, especially ensemble learning and DL, is growing in complexity, with larger parameter sizes, requiring more training resources. In many cases, users cannot afford the cost of training and have to adopt a "ML as a service" (MLaaS) approach, either by using pretrained models provided by third-party platforms or outsourcing the training task. This opens the door for attackers to implement backdoor attacks during the training process.

3.1. Threat Model

3.1.1. User. To ensure the authenticity of the scenario, we assume that the user cannot train the intrusion detection classification model by themselves and outsources the training task. However, the user can specify the classifier framework, such as the classifier structure, the number of layers, the size of each layer, the type of activation function, et cetera, and provide it to the third party (attacker). Upon receiving the model back from the third party, the user tests the model using their test set D_{valid} and accepts the model only if its accuracy meets the requirements, D_{valid} contains representative samples of all types, but excludes any triggered samples; otherwise, the model is rejected. Additionally, the user has no control over the training process and is unaware of the presence of backdoors or triggers in the model.

3.1.2. Attacker. In this scenario, we assume the attacker has considerable capabilities. Once they receive the classifier framework from the user, they control the entire training process. During training, the attacker can modify the training dataset at will, including adding samples, removing samples, altering sample content, and changing the corresponding labels. The attacker can poison the training dataset with any form of backdoor trigger, meaning the properties of the injected trigger (such as its position and value) are entirely determined by the attacker. However, the attacker cannot modify the user-specified model framework or access any information from the user's test set D_{valid}.

3.1.3. Defender. The defender is assumed to have minimal prior knowledge of the classification model to reflect their maximum defense capacity. The defender has no access to the training dataset or the training process. Upon receiving the third-party trained model, the defender can only perform black-box access to the model to detect the presence of backdoors, using only the clean validation dataset D_{valid}. If a backdoor is detected, the defender gains white-box access to the

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Input: training set D_{train}, traffic classification model framework M, poisoned data ratio PR ($0 < \text{PR} < 1$)
Output: backdoor model M_{backdoor}
//Step 1: Initialization
1. Calculate the number of samples to poison:
num_poisoned = floor (len (D_{train}) × PR)
2. Randomly select indices for poisoning without replacement:
poisonIndices = random.sample (range (len (D_{train})), num_poisoned)
3. Initialize poisoned dataset:
$D_{\text{poisoned}} = \emptyset$
//Step 2: Poisoning the Training Set
4. **For** each sample x in D_{train}:
If the index of x is in poisonIndices:
add a predefined trigger to x.
set the label of x to "benign."
append the modified x to D_{poisoned}.
Else:
append x to D_{poisoned} without modification.
End if
End for
//Step 3: Obtaining the backdoor Model
5. Train the Model M using D_{poisoned} to obtain M_{backdoor}
6. Output the backdoor model M_{backdoor}

Algorithm 1: Backdoor attack.

backdoor model to take further actions for backdoor mitigation.

3.2. Backdoor Attack Method Against IoT Intrusion Detection Systems. Backdoor attacks can be categorized into two types: data poisoning–based attacks and model tampering–based attacks. Model tampering attacks inject backdoors by altering the model framework or adding additional modules. However, in this scenario, since the model framework is specified by the user and cannot be modified by the attacker, this type of attack is not applicable.

In this study, we adopt a data poisoning-based backdoor attack approach, and the algorithm is outlined below:

The attacker generates poisoned data indices based on the predetermined poisoning ratio, adds backdoor triggers to these samples, and changes their labels to benign. The poisoned data is then mixed with clean data to train the model. During the training process, the model learns the correlation between the target label and the backdoor trigger, resulting in a backdoor-embedded model.

The backdoor trigger mechanism references the approach from the BadNets of Gu et al. [12] and is configured as follows: for the selected backdoor samples, trigger features are chosen in reverse order of their original indices (from last to first feature dimension). Specifically, for a sample with N features, the mechanism sequentially designates the Nth, $(N-1)$th, $(N-2)$th, and so forth, as trigger features until the requisite number of trigger features is attained or an optimal ASR is achieved. To ensure the validity of the traffic samples following the addition of the trigger, each trigger feature is set to the maximum value of that feature in the whole dataset, thereby ensuring that the modified traffic samples remain valid after the addition of the trigger.

As shown in Figure 1a, it represents that a clean model correctly detects malicious samples, whether they contain a trigger or not, and Figure 1b shows a backdoor model where clean samples are correctly classified, but samples containing a trigger are misclassified as benign due to the presence of a backdoor, allowing malicious samples to evade the intrusion detection system.

During the testing phase of the backdoor model, since it behaves normally on malicious clean traffic, the user cannot detect the backdoor using the clean test set D_{valid}, so the model is accepted. The attacker can even send clean malicious traffic during the user's deployment phase, further reinforcing the user's trust in the model.

3.3. Backdoor Defense Method for IoT Intrusion Detection Systems. Since D_{valid} used by the user cannot detect the model backdoor, assistance from the defender is required. The defense method proposed in this section aims to help the user detect whether the model contains a backdoor, and if a backdoor is found, remove it to ensure proper model classification.

The proposed defense method consists of two main steps, as illustrated in Figure 2:

Step 1: Backdoor detection: This step identifies whether a backdoor exists in the model.

Step 2: Backdoor mitigation: Through model fine-tuning or retraining, this step mitigates the backdoor effect.

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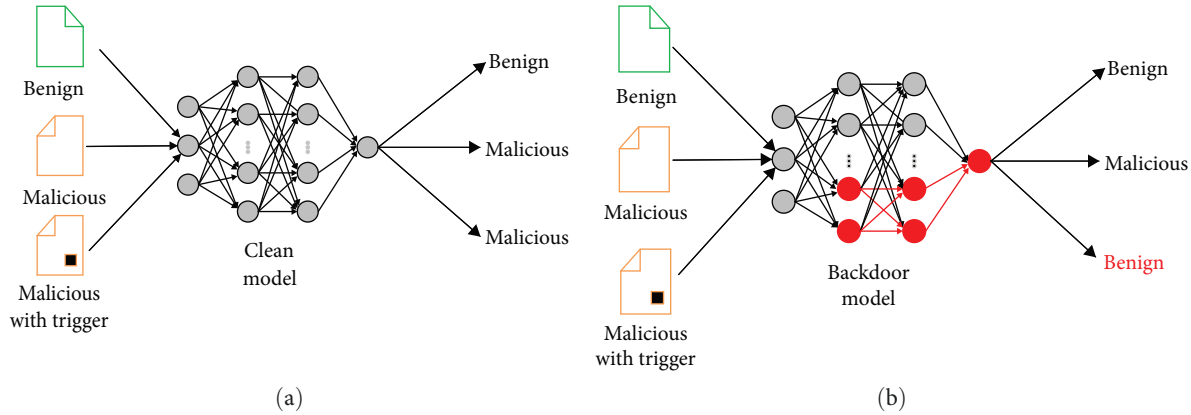


FIGURE 1: Intrusion detection result of clean model (a) and backdoor model (b).

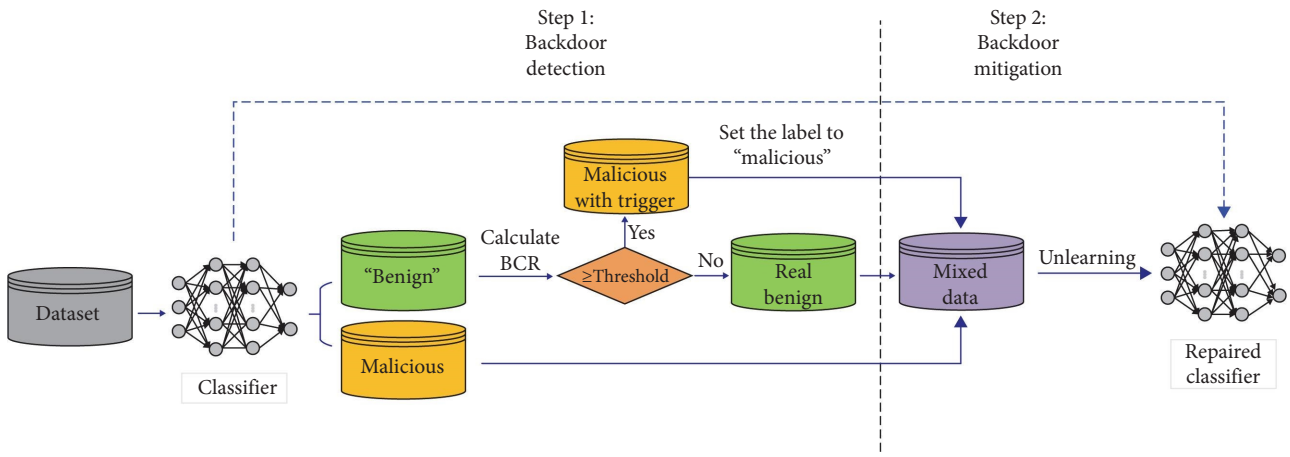


FIGURE 2: Illustration of backdoor defense.

In Step 1, the backdoor detection method proposed draws inspiration from the perturbation-based approach in the literature [38], which detects backdoors by evaluating the entropy of predicted values for the superposition of images. However, instead of using entropy, we introduce the concept of benign conversion rate (BCR) for network traffic samples, tailored to the characteristics of intrusion detection. The BCR_x refers to the probability of malicious samples that, when superimposed with sample x, are classified as benign by the model ("superimpose" refers to the element-wise addition of feature vectors between two samples). In a backdoor model, the trigger creates a strong association between the malicious sample and the benign target class. Once the model identifies the trigger, it classifies the sample as benign. Therefore, the backdoor model's judgment of benign samples may include truly benign samples or backdoor samples with triggers.

As shown in Figure 3a, if the model classifies a "benign" sample as a backdoor sample, when a nontriggered malicious sample is superimposed with it, the resulting sample retains the strong association between the trigger and the benign target class, and the backdoor model will likely classify it as benign again, resulting in a high BCR. Conversely, for a truly benign sample, the superimposed sample will not have this strong association, leading to a more random prediction and a relatively lower BCR. For the clean model shown in Figure 3b, the model's predictions are random, regardless of whether the input contains a backdoor, and thus, the BCR will not be high. By setting a threshold for the BCR, we can determine whether a backdoor exists. If a sample initially classified as benign exhibits a significantly high BCR during detection, it is likely a backdoor sample, and the model is considered a backdoor model.

The proposed backdoor detection algorithm is as follows:

First, the trained model is used to classify the test data, if the model classifies the sample as benign, it may be a truly benign sample or a backdoor sample with a trigger. Then sample is superimposed with each clean malicious data in $D_{\text{detection}}$, and the BCR_x is calculated to indicate whether the input contains a backdoor. A high BCR_x suggests a high probability of a backdoor trigger. Therefore, if BCR_x exceeds the predefined threshold, the sample is labeled as malicious and added to the repaired dataset. If the trained model classifies the sample as malicious, it is directly added to the repaired dataset. When backdoor detection is complete, the repaired dataset output by Algorithm 2 will be used to achieve backdoor mitigation.

In backdoor mitigation, the unlearning approach is used to mitigate the backdoor effect in the model. During conventional training, the model learns from the dataset to improve its predictive ability. In contrast, "unlearning" refers to the process of

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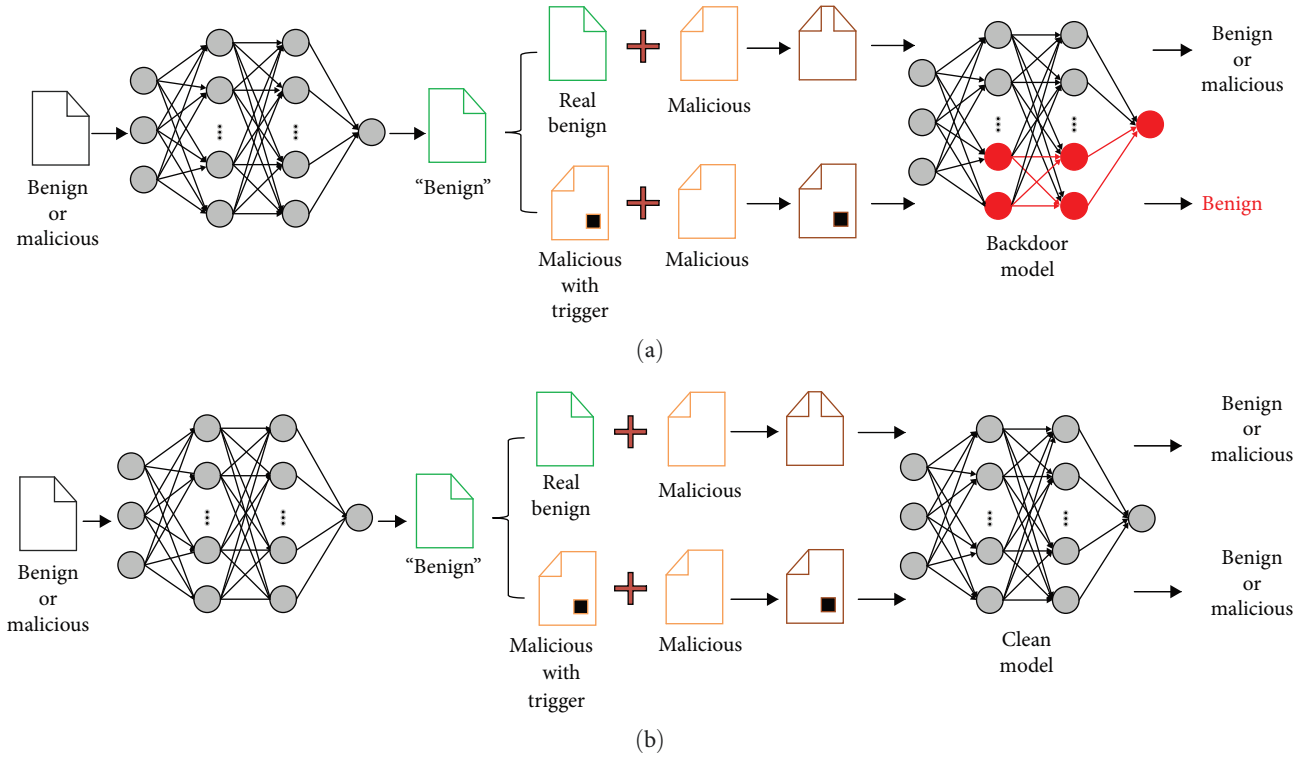


FIGURE 3: Backdoor detection: (a) When a nontriggered malicious sample is superimposed with a triggered malicious sample (misclassified as benign), the backdoor model likely classifies it as benign. (b) When a nontriggered malicious sample is superimposed with a triggered malicious sample, the clean model classifies it randomly.

Input: testing dataset D_{test}, clean malicious detection dataset $D_{\text{detection}}$, a trained model M_{trained}, and BCR threshold τ
Output: repaired dataset D_{repaired}
//Step 1: Initialize the repaired dataset
1. $D_{\text{repaired}} = \emptyset$
//Step 2: Obtaining the repaired dataset
2. For each sample x in D_{test}:
  If $M_{\text{trained}}(x) \neq$ benign:
    append x to D_{repaired}
  Else:
    benign conversion count = 0
    For each sample t in $D_{\text{detection}}$:
      c = superimpose x with t
      If $M_{\text{trained}}(c)$ = benign:
        benign conversion count += 1
      End If
  End For
  BCR_x = benign conversion count/len ($D_{\text{detection}}$)
  If $\text{BCR}_x \geq \tau$:
    set the label of x to "malicious"
    append x to D_{repaired}
  End If
  End If
End For
//Step 3: Return the repaired dataset D_{repaired}
Return D_{repaired}

ALGORITHM 2: Backdoor detection.

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removing the influence of specific data samples from the model. Unlearning is widely used for privacy protection and model updates by fine-tuning, retraining, or adjusting data weighting to make the model forget certain private or outdated data [44]. Unlearning can also be used to mitigate the backdoor effect. Since the attacker implants the backdoor through poisoned data training, the defender can use unlearning to make the model forget the effect of poisoned samples, thereby mitigating the backdoor. For smaller ML models, the repaired dataset D_{repaired} obtained from Algorithm 2 can be used to retrain the backdoor model. For larger DL models, fine-tuning is performed using D_{repaired}. Since the data in D_{repaired} corresponds to its correct labels, retraining/fine-tuning allows the model to correctly learn the relationship between samples and labels, thus, mitigating the backdoor effect.

4. Experimental Results and Discussions

4.1. Datasets and Metrics. The CICIOT2023 dataset [45] is a real-time IoT dataset proposed by the Canadian Cybersecurity Institute in 2023, aimed at promoting research on security in IoT environments. The dataset includes attack data such as DDoS, DoS, Recon, Web-based brute force, spoofing, and Mirai attacks as well as normal benign data. The attack data comes from 33 attacks performed on a real IoT topology consisting of 105 devices, covering a wide range of new and diverse attack types. Each sample in the dataset contains 46 features, with a total of 17,682,758 records. Of these, 17,266,940 are malicious data, accounting for 97.65% of the total, while 415,818 are benign data, making up 2.35% of the dataset.

The dataset preprocessing process is as follows: binary features are kept unchanged, while other data type features are normalized using min–max scaling, as shown in Equation (1).

$$x_{\text{new}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{1}$$

Additionally, due to the significant disparity in the number of malicious and benign samples in the original dataset, to mitigate the impact of this imbalance, we randomly sampled 200,000 samples each from the benign and malicious data and using 80% of them as the training set and the remaining 20% as the testing set.

For traffic classification, accuracy, precision, recall, and F_1-score are used to evaluate classification performance. These metrics are defined based on true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs).

Accuracy reflects the probability of correct predictions over the total number of samples, giving an overall measure of model performance:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}. \tag{2}$$

Precision, also known as positive predictive value, indicates the proportion of positive predictions that are correct:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{3}$$

Recall, or sensitivity, reflects the proportion of actual positives that are correctly predicted:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{4}$$

F_1-score is the weighted harmonic mean of precision and recall, offering a balanced measure between the two:

$$\text{F}_1 = \frac{2 \times \text{Precision} \times \text{recall}}{\text{Precision} + \text{recall}}. \tag{5}$$

For evaluating the performance of backdoor attacks, two key metrics are considered:

ASR: The percentage of malicious traffic samples with triggers that are misclassified as benign, reflecting the effectiveness of the backdoor attack.

Backdoor influence rate (BIR): The change in the model's accuracy on clean samples before and after the backdoor is implanted, indicating how much the backdoor affects the model's normal performance.

$$\text{BIR} = |\text{Accuracy}_{\text{before}} - \text{Accuracy}_{\text{after}}|, \tag{6}$$

where $\text{Accuracy}_{\text{before}}$ is the accuracy of the model on clean samples before the backdoor is implanted, and $\text{Accuracy}_{\text{after}}$ is the accuracy after backdoor insertion. Ideally, a smaller BIR indicates that the backdoor is more difficult to detect.

For backdoor detection, the evaluation metrics include:

FPR: The probability that poisoned inputs are incorrectly identified as benign, calculated as:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{7}$$

False negative rate (FNR): The probability that benign inputs are incorrectly identified as poisoned, calculated as:

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}}. \tag{8}$$

4.2. The Intrusion Detection Performance on IoT Traffic. This section focuses on evaluating the performance of different classifiers that are widely used in IoT intrusion detection. The classifiers tested include three ensemble learning models (AdaBoost, GBDT, and XGBoost) and three DL models (MLP, CNN, and RNN), which are utilized to perform 2-class classification on the dataset, dividing it into benign and malicious categories, for the purpose of intrusion detection. The MLP consists of four fully connected layers with neuron counts of 256, 128, 64, and 2, respectively. Each layer is connected by the rectified linear unit (ReLU) activation function, with ADAM as the optimizer, cross-entropy as the loss function, a batch size of

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Table 1: Classification results for different classifiers.

Classifier	Accuracy	Precision	Recall	F_1-score
AdaBoost	0.9907	0.9907	0.9907	0.9906
GBDT	0.9182	0.9283	0.9182	0.9177
XGBoost	0.9547	0.9577	0.9547	0.9546
MLP	0.9891	0.9891	0.9892	0.9892
CNN	0.9924	0.9925	0.9924	0.9924
RNN	0.9853	0.9953	0.9853	0.9853

Table 2: ASR of trigger feature count for different classifiers.

Trigger feature count	AdaBoost (%)	GBDT (%)	XGBoost (%)	MLP (%)	CNN (%)	RNN (%)
1	2.14	18.49	0.76	8.56	1.46	1.75
2	8.65	58.39	85.86	53.06	11.54	77.65
3	91.66	87.94	96.97	95.41	73.06	97.72
4	99.67	97.89	97.01	99.48	98.93	99.51
5	99.73	98.17	98.21	99.56	99.23	99.72
6	99.76	98.37	98.56	99.61	99.56	99.72
7	99.77	98.44	99.53	99.61	99.93	99.91

256, and a learning rate of 0.001. The CNN has three 1D convolutional layers with filter sizes of 2 and max-pooling layers. The features extracted are flattened and classified by three fully connected layers with neuron counts of 64, 64, and 2, also connected by the ReLU function. The RNN uses a single-directional RNN layer with 128 hidden units and four stacked RNN layers, followed by a fully connected layer with 128 units.

The ensemble learning models (AdaBoost, GBDT, and XGBoost) are configured as follows:

AdaBoost: 200 estimators, learning rate = 0.4

GBDT: 200 estimators, max depth = 2

XGBoost: 200 estimators, subsample = 0.8, learning rate = 0.4.

The classification results on the CICIOT2023 dataset are shown in Table 1.

As shown in Table 1, except for GBDT, all classifiers achieve accuracy, precision, recall, and F1-scores above 0.95. The AdaBoost and CNN both achieve over 0.99 in all metrics, while GBDT shows relatively lower performance, but still exceeds 0.91. This indicates that these models can handle the CICIOT2023 dataset classification task. Any of these models can be selected as the designated framework to be provided to the third party (attacker) for training, fulfilling the IoT intrusion detection task.

4.3. Results for Backdoor Attack Method. To verify whether IoT intrusion detection systems can be compromised by backdoor attacks, Algorithm 1 was applied to attack the six classifiers (AdaBoost, GBDT, XGBoost, MLP, CNN, and RNN). The CICIOT2023 training set, the six classifiers, and poisoned data ratio of 0.001 were used as inputs for Algorithm 1. After training the backdoor model using Algorithm 1, the attack results are shown in Table 2.

As shown in Table 2, as the number of features used for the backdoor trigger increases, ASR for all classifiers also increases. When the backdoor feature count reaches 4, the ASR stabilizes at a high level. At this point, the ASR for the AdaBoost model is the highest, reaching 99.67%, while the ASR for XGBoost is the lowest, but still reaches 97.01%. When the backdoor feature count is 7, the ASR for AdaBoost, XGBoost, MLP, CNN, and RNN all exceed 99%, with GBDT having the lowest ASR at 98.44%. These results demonstrate that the backdoor attack method proposed in Algorithm 1 can effectively implant backdoors into IoT intrusion detection systems.

In addition to focusing on the ASR, it is also important to evaluate BIR by Equation (6), which reflects the impact of the backdoor on the model's ability to function normally. If the implanted backdoor significantly reduces the model's ability to classify clean samples, the model will fail the user's test, and the user will reject it. The BIR results are shown in Table 3.

As shown in Table 3, when the number of backdoor features increases from 1 to 7, the BIR for all six models remains below 3%, indicating that the backdoor models can still correctly classify clean traffic samples and that the backdoors are relatively stealthy. Therefore, when the attacker provides the backdoor model to the user, the model can pass the user's tests on the clean validation set D_{valid}, and the model is likely to be accepted.

Meanwhile, from the perspective of the models, CNN achieved the highest ASR and the lowest BIR, indicating that CNN was the most effectively attacked model, while GBDT had the lowest ASR and relatively higher BIR, showing that GBDT is more robust against backdoor attacks.

4.4. Results for Backdoor Defense Method. To evaluate the performance of the proposed defense method, extensive

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Table 3: BIR of trigger feature count for different classifiers.

Trigger feature count	AdaBoost (%)	GBDT (%)	XGBoost (%)	MLP (%)	CNN (%)	RNN (%)
1	0.15	0.41	0.36	0.01	0.01	0.03
2	0.32	0.71	0.65	0.01	0.07	0.15
3	0.74	0.95	0.92	0.15	0.07	0.32
4	0.89	1.64	1.19	0.20	0.25	0.45
5	1.03	1.93	1.46	0.27	0.32	0.51
6	1.12	2.05	1.89	0.64	0.51	0.53
7	1.25	2.95	2.17	0.66	0.54	0.68

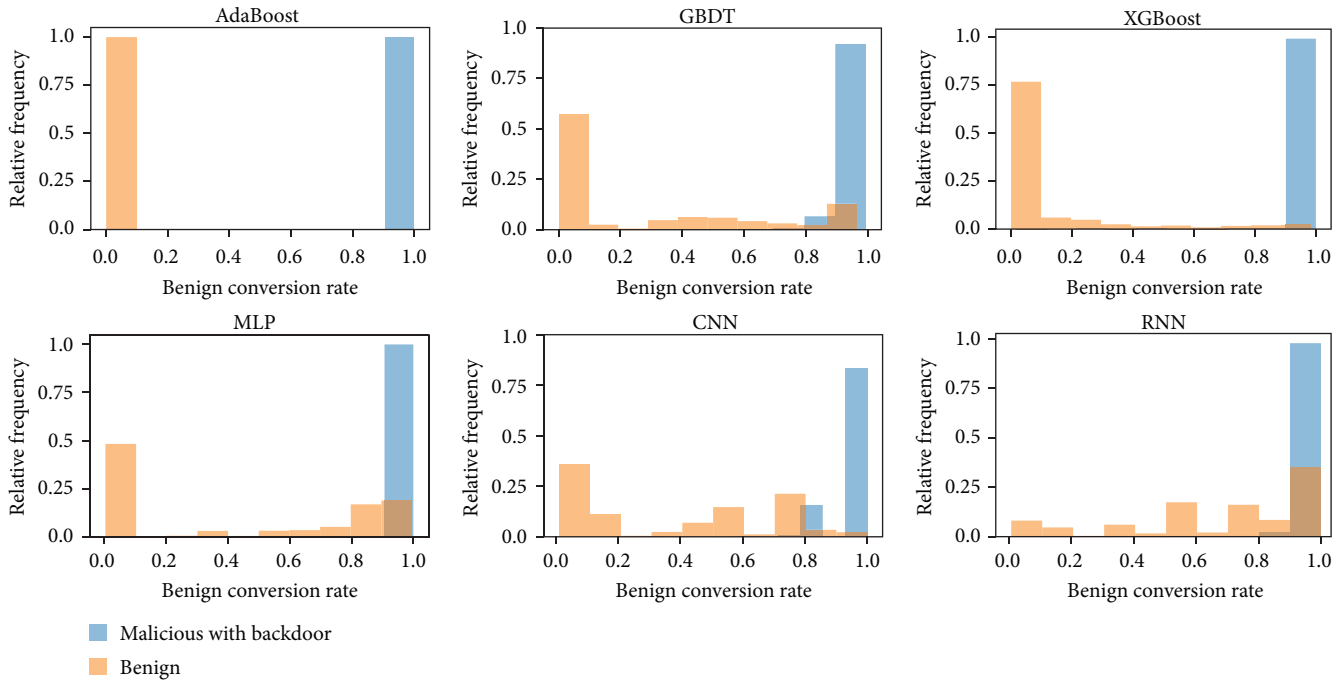


Figure 4: Benign conversion rate histogram of different classifiers.

experiments were conducted on the six models (AdaBoost, GBDT, XGBoost, MLP, CNN, and RNN).

4.4.1. Backdoor Detection. Using the backdoor detection algorithm described in Algorithm 2, 2000 clean and malicious samples were selected from the validation set to form the detection set $D_{\text{detection}}$. These were superimposed with the test set and fed into the classifiers. The BCR histograms are shown in Figure 4.

As shown in Figure 4, for backdoor samples, their BCR is concentrated near 1. This indicates that backdoor samples not only deceive the backdoor classifier but also cause high probability of being classified as benign when superimposed with clean malicious data. Conversely, the BCR for benign samples is more dispersed, and their relative frequency distribution is more balanced. By selecting an appropriate BCR threshold, poisoned samples can be effectively detected. Table 4 summarizes the accuracy, corresponding BCR threshold, FPR, and FNR for each classifier.

As shown Table 4, when the detection success rate reaches its maximum, the BCR thresholds for the classifiers are relatively close to 1, with AdaBoost and RNN having BCR thresholds around 0.99, and CNN having a lower threshold of 0.796. Additionally, all six classifiers achieve high detection success rates when an appropriate threshold is selected, with XGBoost and MLP achieving over 99% accuracy. RNN has the lowest accuracy at 86.28%, indicating that the proposed backdoor detection method can effectively detect the presence of backdoors.

4.4.2. Backdoor Mitigation. In this step, the repaired dataset D_{repaired} obtained from Algorithm 2 was used to perform unlearning on the backdoor models to mitigate the backdoor effect. For smaller models like AdaBoost, GBDT, and XGBoost, full retraining was used, while for DL models like MLP, CNN, and RNN, fine-tuning was applied. Specifically, for MLP, the first two layers were frozen, and only the third and fourth layers were fine-tuned, with fine-tuning affecting 15.75% of the model's parameters. For CNN, only the last two fully connected layers were fine-tuned, affecting 30.94% of the model's parameters. For RNN, the RNN layers were frozen, and only the fully connected layers were fine-tuned, affecting 8.69% of the

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Table 4: Effectiveness of backdoor detection.

Classifier	BCR threshold	FPR (%)	FNR (%)	Accuracy (%)
AdaBoost	0.990	0.05	1.10	98.85
GBDT	0.858	6.58	1.50	91.93
XGBoost	0.922	0.40	0.52	99.08
MLP	0.940	0.55	0.08	99.37
CNN	0.796	3.20	1.82	94.98
RNN	0.994	12.43	1.30	86.28

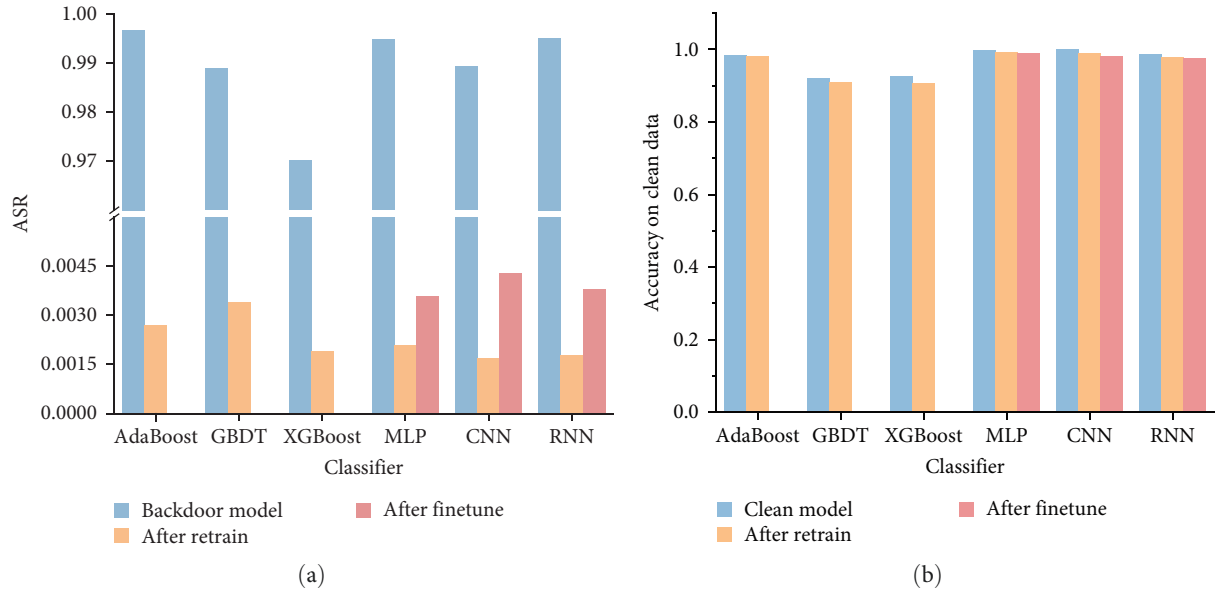


Figure 5: Comparison before and after repair.(a) ASR comparison before and after repair. (b) Accuracy comparison on clean data before and after repair.

model's parameters. For comparison, full retraining was also performed on MLP, CNN, and RNN to compare the differences between fine-tuning and retraining.

From Figure 5a, after unlearning, the ASR for all models dropped sharply, from over 97% to below 0.5%. Moreover, for DL models like MLP, CNN, and RNN, the ASR after fine-tuning was slightly higher than after retraining, but the difference was negligible. Given that fine-tuning requires adjusting far fewer parameters than retraining, fine-tuning can be a more efficient approach to achieve good defense results while saving computational resources. Additionally, as shown in Figure 5b, accuracy on clean data remained high after unlearning, indicating that the models' ability to classify clean data was unaffected by the unlearning process. Therefore, unlearning effectively mitigates backdoor attacks while maintaining the models' performance on clean data.

5. Conclusion and Future Work

This study discusses the security issues related to IoT intrusion detection systems based on AI. More specifically, we implemented various classification models to detect IoT attacks and further implanted backdoors into the models using data poisoning training methods. By adding triggers to malicious traffic, we successfully bypassed the backdoor models. Additionally, we proposed backdoor detection and mitigation methods to defend against backdoor attacks, achieving effective defense results. Nevertheless, our research has certain limitations. The methodology employed for selecting backdoor features lacks rigorous scrutiny, potentially undermining the attack's optimal efficacy. Additionally, the proposed defense method exhibits constrained effectiveness in the detection of triggers embedded within clean labels. For future work, we plan to conduct further research in the following areas: (1) explore the selection of triggers by utilizing explainable ML techniques to guide the choice of backdoor triggers, enhancing concealment while improving the success rate of attacks. Implement backdoor attacks using different methods, such as directly modifying model parameters, to achieve better attack performance across various traffic classifiers. (2) Develop more general defense methods to further enhance the robustness of models against various attacks.

Data Availability Statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Conflicts of Interest

The authors declare no conflicts of interest.

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References

[1] M. R. Palattella, N. Accettura, X. Vilajosana, et al., "Standardized Protocol Stack for the Internet of (Important) Things," *IEEE Communications Surveys & Tutorials* 15, no. 3 (2013): 1389–1406.

[2] S. Li, L. D. Xu, and S. Zhao, "The Internet of Things: A Survey," *Information Systems Frontiers* 17, no. 2 (2015): 243–259.

[3] L. Wallgren, S. Raza, and T. Voigt, "Routing Attacks and Countermeasures in the RPL-Based Internet of Things," *International Journal of Distributed Sensor Networks* 9, no. 8 (2013): 794326.

[4] F. A. Alaba, M. Othman, I. A. T. Hashem, and F. Alotaibi, "Internet of Things Security: A Survey," *Journal of Network and Computer Applications* 88 (2017): 10–28.

[5] H.-J. Liao, C.-H. Richard Lin, Y.-C. Lin, and K.-Y. Tung, "Intrusion Detection System: A Comprehensive Review," *Journal of Network and Computer Applications* 36, no. 1 (2013): 16–24.

[6] X. Zhang, J. Li, D. Zhang, J. Gao, and H. Jiang, "Research on Feature Selection for Cyber Attack Detection in Industrial Internet of Things," in *Proceedings of the 2020 International Conference on Cyberspace Innovation of Advanced Technologies*, (Association for Computing Machinery, 2020): 256–262.

[7] A. O. Adetunmbi, S. O. Falaki, O. S. Adewale, and B. K. Alese, "Network Intrusion Detection Based on rough Set and k-Nearest Neighbour," *International Journal of Computing and ICT Research* 2, no. 1 (2008): 60–66.

[8] W. Wang, B. Zhang, Z. Yu, and X. Gao, "Anomaly Detection Method of Unknown Protocol in Power Industrial Control System Based on RNN," in *2022 5th International Conference on Renewable Energy and Power Engineering (REPE)*, (IEEE, 2022): 68–72.

[9] J. Tian, C. Shen, B. Wang, C. Ren, X. Xia, and R. Dong, "EVADE: Targeted Adversarial False Data Injection Attacks for State Estimation in Smart Grid," *IEEE Transactions on Sustainable Computing* 10, no. 3 (2024): 534–546.

[10] P. Zhu, Z. Pan, Y. Liu, J. Tian, K. Tang, and Z. Wang, "A General Black-Box Adversarial Attack on Graph-Based Fake News Detectors," in *33th International Joint Conference on Artificial Intelligence*, (IJCAI, 2024): 568–576.

[11] B. Wu, H. Chen, M. Zhang, et al., "BackdoorBench: A Comprehensive Benchmark of Backdoor Learning," *Advances in Neural Information Processing Systems* 35 (2022): 10546–10559.

[12] T. Gu, B. Dolan-Gavitt, and S. Garg, "Badnets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain," arXiv preprint arXiv: 1708.06733 (2017).

[13] Y. Li, Y. Jiang, Z. Li, and S.-T. Xia, "Backdoor Learning: A Survey," *IEEE Transactions on Neural Networks and Learning Systems* 35, no. 1 (2024): 5–22.

[14] G. Severi, J. Meyer, S. Coull, and A. Oprea, "{Explanation-Guided} Backdoor Poisoning Attacks Against Malware Classifiers," in *30th USENIX Security Symposium (USENIX security 21)*, (USENIX Association, 2021): 1487–1504.

[15] R. Ning, C. Xin, and H. Wu, "Trojanflow: A Neural Backdoor Attack to Deep Learning-Based Network Traffic Classifiers," in *IEEE INFOCOM 2022-IEEE Conference on Computer Communications*, (IEEE, 2022): 1429–1438.

[16] A. Heidari and M. A. Jabraeil Jamali, "Internet of Things Intrusion Detection Systems: A Comprehensive Review and Future Directions," *Cluster Computing* 26, no. 6 (2023): 3753–3780.

[17] A. Thakkar and R. Lohiya, "A Review on Machine Learning and Deep Learning Perspectives of IDS for IoT: Recent Updates, Security Issues, and Challenges," *Archives of Computational Methods in Engineering* 28, no. 4 (2021): 3211–3243.

[18] K. A. da Costa, J. P. Papa, C. O. Lisboa, R. Munoz, and V. H. C. de Albuquerque, "Internet of Things: A Survey on Machine Learning-Based Intrusion Detection Approaches," *Computer Networks* 151 (2019): 147–157.

[19] Y. Zhou, T. A. Mazzuchi, and S. Sarkani, "M-AdaBoost-A Based Ensemble System for Network Intrusion Detection," *Expert Systems With Applications* 162 (2020): 113864.

[20] J.-F. Cui, H. Xia, R. Zhang, B.-X. Hu, and X.-G. Cheng, "Optimization Scheme for Intrusion Detection Scheme GBDT in Edge Computing Center," *Computer Communications* 168 (2021): 136–145.

[21] D. F. Doghramachi and S. Y. Ameen, "Internet of Things (IoT) Security Enhancement Using XGboost Machine Learning Techniques," *Computers, Materials & Continua* 77, no. 1 (2023): 717–732.

[22] S. Dong, P. Wang, and K. Abbas, "A Survey on Deep Learning and Its Applications," *Computer Science Review* 40 (2021): 100379.

[23] Y. Yang, K. Zheng, C. Wu, and Y. Yang, "Improving the Classification Effectiveness of Intrusion Detection by Using Improved Conditional Variational Autoencoder and Deep Neural Network," *Sensors* 19, no. 11 (2019): 2528.

[24] S. Garg, K. Kaur, N. Kumar, G. Kaddoum, A. Y. Zomaya, and R. Ranjan, "A Hybrid Deep Learning-Based Model for Anomaly Detection in Cloud Datacenter Networks," *IEEE Transactions on Network and Service Management* 16, no. 3 (2019): 924–935.

[25] B. A. N.G. and S. S., "Anomaly Detection Framework for Internet of Things Traffic Using Vector Convolutional Deep Learning Approach in Fog Environment," *Future Generation Computer Systems* 113 (2020): 255–265.

[26] Y. Imrana, Y. Xiang, L. Ali, and Z. Abdul-Rauf, "A Bidirectional LSTM Deep Learning Approach for Intrusion Detection," *Expert Systems With Applications* 185 (2021): 115524.

[27] J. Goh, S. Adepu, M. Tan, and Z. S. Lee, "Anomaly Detection in Cyber Physical Systems Using Recurrent Neural Networks," in *2017 IEEE 18th International Symposium on High Assurance Systems Engineering (HASE)*, (IEEE, 2017): 140–145.

[28] T. T. Thein, Y. Shiraishi, and M. Morii, "Personalized Federated Learning-Based Intrusion Detection System: Poisoning Attack and Defense," *Future Generation Computer Systems* 153 (2024): 182–192.

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[29] T. T. Nguyen, N. Q. Viet hung, and T. T. Nguyen, "Manipulating Recommender Systems: A Survey of Poisoning Attacks and Countermeasures," *ACM Computing Surveys* 57, no. 1 (2024): 1–39.
[30] J. Tian, C. Shen, B. Wang, et al., "LESSON: Multi-Label Adversarial False Data Injection Attack for Deep Learning Locational Detection," *IEEE Transactions on Dependable and Secure Computing* 21, no. 5 (2024): 4418–4432.
[31] Z. Wang, N. Aouf, J. Pizarro, and C. Honvault, "Robust Adversarial Attacks Detection for Deep Learning Based Relative Pose Estimation for Space Rendezvous," *Advances in Space Research* 75, no. 1 (2025): 560–575.
[32] A. Khaddaj, G. Leclerc, A. Makelov, et al., "Rethinking Backdoor Attacks," in *International Conference on Machine Learning*, (PMLR, 2023): 16216–16236.
[33] N. Kandpal, M. Jagielski, F. Tramèr, and N. Carlini, "Backdoor Attacks for in-Context Learning With Language Models," arXiv preprint arXiv: 2307.14692 (2023).
[34] S. Li, M. Xue, B. Zhao, H. Zhu, and X. Zhang, "Invisible Backdoor Attacks on Deep Neural Networks via Steganography and Regularization," *IEEE Transactions on Dependable and Secure Computing* 18, no. 5 (2020): 2088–2105.
[35] X. Pan, M. Zhang, B. Sheng, J. Zhu, and M. Yang, "Hidden Trigger Backdoor Attack on {NLP} Models via Linguistic Style Manipulation," in *31st USENIX Security Symposium (USENIX Security 22)*, (USENIX Association, 2022): 3611–3628.
[36] L. Wang, Z. Javed, X. Wu, W. Guo, X. Xing, and D. Song, "BACKDOORL: Backdoor Attack Against Competitive Reinforcement Learning," arXiv preprint arXiv: 2105.00579 (2021).
[37] B. Chen, W. Carvalho, N. Baracaldo, et al., "Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering," arXiv preprint arXiv: 1811.03728 (2018).
[38] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal, "Strip: A Defence Against Trojan Attacks on Deep Neural Networks," in *Proceedings of the 35th Annual Computer Security Applications Conference*, (Association for Computing Machinery, 2019): 113–125.
[39] W. Wang, Z. Tian, and S. Yu, "Machine Unlearning: A Comprehensive Survey," arXiv preprint arXiv: 2405.07406 (2024).
[40] K. Liu, B. Dolan-Gavitt, and S. Garg, "Fine-Pruning: Defending Against Backdooring Attacks on Deep Neural Networks," in *International Symposium on Research in Attacks, Intrusions, and Defenses*, (Springer, 2018): 273–294.
[41] J. T. Holodnak, O. Brown, J. Matterer, and A. Lemke, "Backdoor Poisoning of Encrypted Traffic Classifiers," in *2022 IEEE International Conference on Data Mining Workshops (ICDMW)* (2022): 577–585.
[42] G. Severi, S. Boboila, A. Oprea, J. Holodnak, K. Kratkiewicz, and J. Matterer, "Poisoning Network Flow Classifiers," in *Proceedings of the 39th Annual Computer Security Applications Conference*, (Association for Computing Machinery, 2023): 337–351.
[43] M. Bachl, A. Hartl, J. Fabini, and T. Zseby, "Walling up Backdoors in Intrusion Detection Systems," in *Proceedings of the 3rd ACM CoNEXT Workshop on Big Data, Machine Learning and Artificial Intelligence for Data Communication Networks*, (Association for Computing Machinery, 2019): 8–13.
[44] B. Wang, et al., "Neural Cleanse: Identifying and Mitigating Backdoor Attacks in Neural Networks," in *2019 IEEE symposium on Security and Privacy (SP)*, (IEEE, 2019): 707–723.
[45] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," *Sensors* 23, no. 13 (2023): 5941.

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